

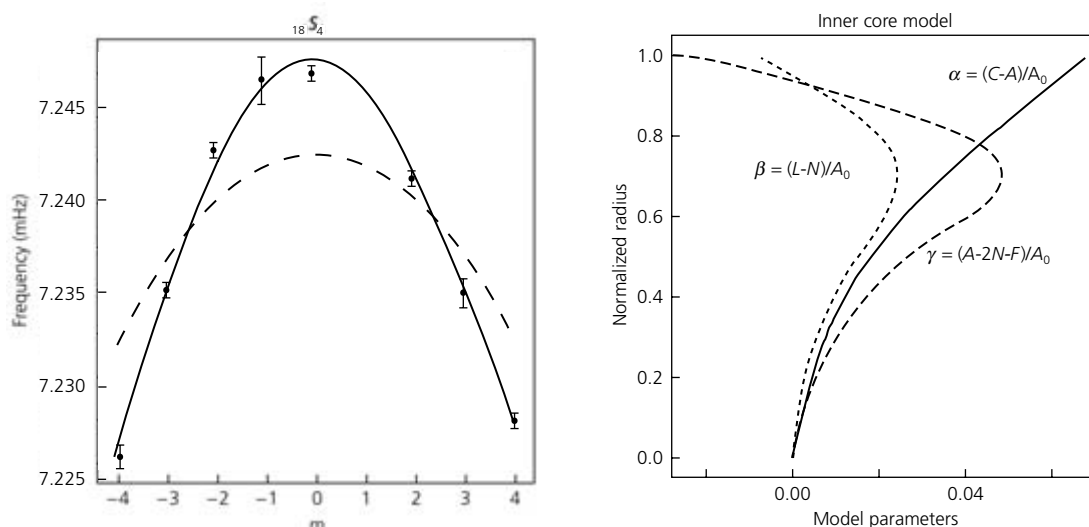


Fig. 3.6-13 Evidence for transverse isotropy in the inner core from the splitting of a spheroidal normal mode multiplet. Points represent the observed frequencies for the different azimuthal orders (*left*). The dashed curve is the prediction for the inner core including the effects of the earth's rotation and ellipticity but not anisotropy. The solid curve incorporates the anelastic model at the *right* by combinations of the elastic constants for transverse isotropy. In this formulation, α , β , and γ differ from their usual seismological definitions. (Tromp, 1993. Reproduced with permission from *Nature*.)

3.7 Attenuation and anelasticity

3.7.1 Wave attenuation

In the last section, we extended our view of the earth as an isotropic elastic medium to include the effects of anisotropy. We now consider *anelasticity*, or deviation from elasticity, which is one of the reasons why seismic waves *attenuate* or decrease in amplitude as they propagate. We have already discussed how the reflection and transmission of seismic waves at discrete interfaces reduce their amplitudes. Here, we consider four other processes that can reduce wave amplitudes: *geometric spreading*, *scattering*, *multipathing*, and anelasticity. The first three are elastic processes, in which the energy in the propagating wave field is conserved. By contrast, anelasticity, sometimes called *intrinsic attenuation*, involves conversion of seismic energy to heat.

As in many seismological applications, it is worth first considering familiar analogous behaviors for light. As you move away from a street lamp at night, the light appears dimmer for several reasons. The first is geometric spreading: light moves outward from the lamp in expanding spherical wave fronts (Section 2.4.3). By the conservation of energy, the energy in a unit area of the growing wave front decreases as r^{-2} , where r is the radius of the sphere or distance from the lamp.

Second, the light dims as it is scattered by air molecules, dust, and water in the air. As we have discussed, scattering results when objects acting as Huygens' sources scatter energy in all directions. This effect is dramatic on a foggy night because the scattered light causes a halo around the lamp.

Third, the light is focused or defocused by changes in the refractive properties of the air.¹ This effect is termed *multipathing* in seismology. Focusing and defocusing can be illustrated by looking at the street light through binoculars. Looking through binoculars the usual way, the waves are focused by the lenses, and the lamp appears closer and brighter. Reversing the binoculars makes the lamp appear further away and dimmer.

Fourth, some of the light energy is absorbed by the air and converted to heat. This process differs from the other three in that light energy is actually lost, not just moved onto a different path.

All four processes are important for seismic waves. The first three are described by elastic wave theory, and can increase or decrease an arrival's amplitude by shifting energy within the wave field. By contrast, anelasticity reduces wave amplitudes only because energy is lost from the elastic waves. So much of seismology is built upon the approximation that the earth responds elastically during seismic propagation that it is easy to forget that the earth is not perfectly elastic. However, without anelasticity, seismic waves from every earthquake that ever occurred would still be reverberating until the accumulating reverberations shattered the earth. Elasticity is a good approximation for the earth's response to seismic waves, but

¹ This process causes mirages, where light is refracted differently by hot air just above the ground. Similarly the distorted appearance of the setting sun results from seeing different parts of it through different levels of the atmosphere which refract light differently because of the vertical density gradient.

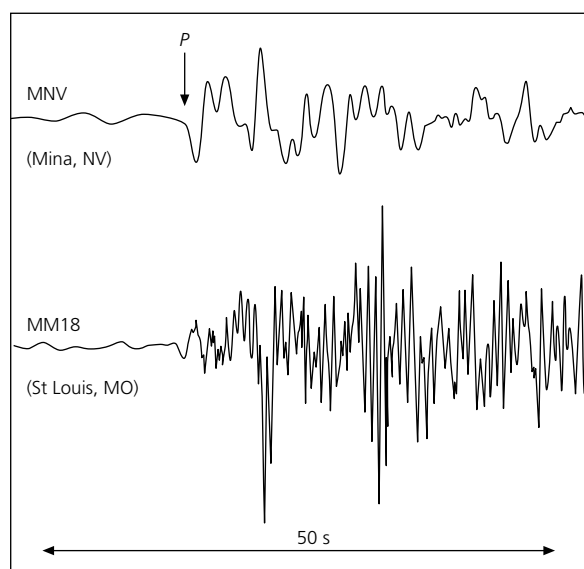


Fig. 3.7-1 Regional variations in attenuation seen in seismograms from an April 14, 1995, earthquake in Texas recorded in Nevada (MNV, $\Delta = 15^\circ$) and Missouri (MM18, $\Delta = 14^\circ$). The MNV record has less high frequency energy because the tectonically active western USA is more attenuating than the stable mid-continent.

there are many important implications and applications of anelasticity.

Anelasticity results because the kinetic energy of elastic wave motion is lost to heat by permanent deformation of the medium. The large-scale, or macroscopic, term for this process is *internal friction*. Among the smaller-scale, or microscopic, mechanisms that may cause this dissipation are stress-induced migration of defects in minerals, frictional sliding on crystal grain boundaries, vibration of dislocations, and the flow of hydrous fluids or magma through grain boundaries. Theoretical and experimental work is being carried out to examine possible mechanisms of seismic attenuation.

The study of anelasticity has lagged behind that of the elastic wave velocities because of the complexities involved in measuring attenuation and understanding its physical causes. Although measuring seismic wave amplitudes is straightforward, they depend on both the source, which is not perfectly known, and all the elastic and anelastic effects anywhere along the paths that the seismic energy traveled between the source and the receiver. Hence it can be hard to distinguish the effects of anelasticity from elastic processes.

This inherent uncertainty is somewhat compensated by the fact that variations in anelasticity are large, as illustrated by comparison of records of an earthquake in Texas at stations in Nevada and Missouri (Fig. 3.7-1). The Nevada seismogram has much less high-frequency energy, showing that the crust in the western USA is much more attenuating than that in the Midwest. By comparison, seismic velocity variations between these areas are generally less than $\pm 10\%$. Even so, because of the difficulties in measuring attenuation, variations in attenua-

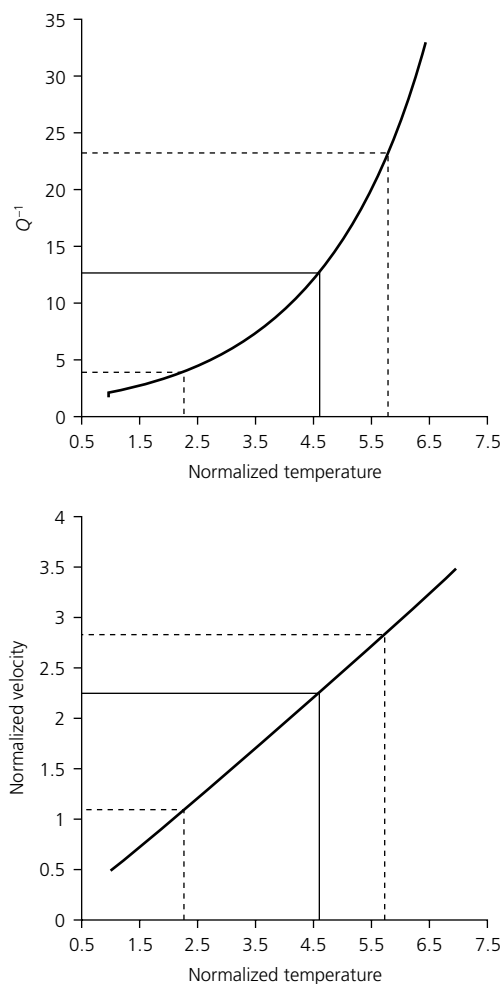


Fig. 3.7-2 Schematic representation of the variations of seismic attenuation (*top*) and normalized velocity (*bottom*) as a function of normalized temperature changes. Attenuation is more sensitive to increased temperature. (Romanowicz, 1995, *J. Geophys. Res.*, 100, 12,375–94, copyright by the American Geophysical Union.)

tion at both regional and global scales are much less resolved than similar variations in velocity.

Attenuation is valuable for studying temperature variations within the earth. Many important geophysical processes (mantle convection, plate tectonics, magmatism, etc.) involve lateral variations in temperature. Elastic velocities are also sensitive to temperature, but are better for mapping cold (fast) anomalies like subducting slabs than hot (slow) material like that at midocean ridges (Section 2.5.10). As shown in Fig. 3.7-2, seismic velocities depend nearly linearly upon temperature, whereas attenuation depends exponentially on temperature. Thus combining velocity and attenuation studies can provide valuable information. Figure 3.7-3 shows the velocity and attenuation structure at a portion of the East Pacific rise axis, where a low-velocity, high-attenuation region is interpreted as a melt-filled magma chamber.

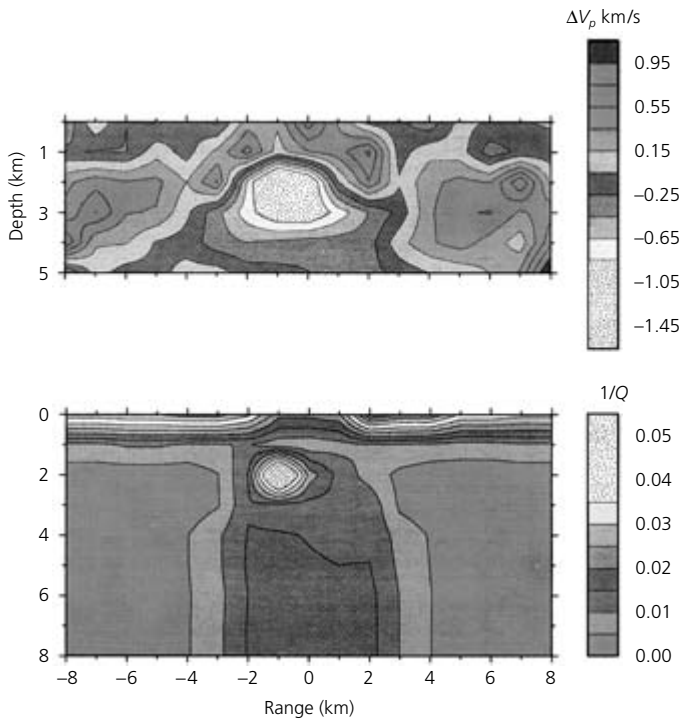


Fig. 3.7-3 Results of *P*-wave velocity (*top*) and attenuation (*bottom*) tomography across the axis of the East Pacific rise. (Solomon and Toomey, 1992, reproduced with the permission of Annual Reviews Inc.)

3.7.2 Geometric spreading

The most obvious effect causing seismic wave amplitudes to vary with distance is geometric spreading, in which the energy per unit wave front varies as a wave front expands or contracts. Geometric spreading differs for surface and body waves. For a homogeneous elastic spherical earth, a surface wave front would spread as it moved from the source to a distance 90° away, refocus as it approached the antipode on the other side of the earth from the source, and so on. The amplitudes would be largest at the source and antipode, where all the energy would be concentrated, and smallest halfway between, 90° from the source. On a homogeneous flat earth, the surface waves would spread out in a growing ring with circumference $2\pi r$, where r is the distance from the source. Conservation of energy² requires that the energy per unit wave front decrease as $1/r$, whereas the amplitudes, which are proportional to the square root of energy (Eqn 2.4.65), decrease as $1/\sqrt{r}$. However, because the earth is a sphere, the ring wraps around the globe (Fig. 3.7-4), making the energy per unit wavefront vary as

$$1/r = 1/(a \sin \Delta), \quad (1)$$

² As we saw in discussing wave reflection and transmission (Section 2.2.4), amplitudes are easier to visualize, but energy is conserved, and hence often more useful for understanding wave behavior.

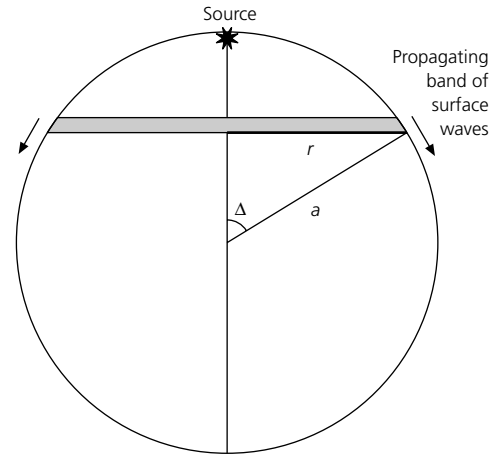


Fig. 3.7-4 Geometric spreading of surface waves for a laterally homogeneous earth yields a wave front that is a ring whose circumference varies as $a \sin \Delta$.

where Δ is the angular distance from the source. Thus the amplitudes decrease as $(a \sin \Delta)^{-1/2}$, with minimum at $\Delta = 90^\circ$, and maxima at 0° and 180° . Actually, not all the energy would focus at the antipode and source even if the earth had no lateral variations in velocity, because some defocusing would result from the earth's ellipsoidal shape. Lateral heterogeneity, discussed next, further distorts the wavefront.

For body waves, consider a spherical wavefront moving away from a deep earthquake. Energy is conserved on the expanding spherical wavefront whose area is $4\pi r^2$, where r is the radius of the wavefront. Thus the energy per unit wave front decays as $1/r^2$, and the amplitude decreases as $1/r$. In reality, because body waves travel through an inhomogeneous earth, their amplitude depends on the focusing and defocusing of rays by the velocity structure. The effects of the variations in velocity with depth were shown in Section 3.4 by considering the density of rays with different incidence angles that arrive at a given distance. These amplitude variations are viewed as geometric spreading and described by the second derivative of the travel time curve (Eqn 3.4.20). Thus, although the phenomenon of geometric spreading is intuitive, quantification of its effects is complicated.

3.7.3 Multipathing

Seismic waves are also focused and defocused by lateral variations in velocity. Although physically this process is the same as the effects of vertical variations, it is often distinguished by the term *multipathing*. The distinction reflects our view of the earth as an essentially layered planet with secondary lateral variations.

As we discussed for tsunamis (Fig. 2.8-9), seismic waves refract towards low-velocity anomalies and away from high-velocity anomalies. Figure 3.7-5 illustrates this effect for a plane wave passing through a refracting layer of variable thickness.

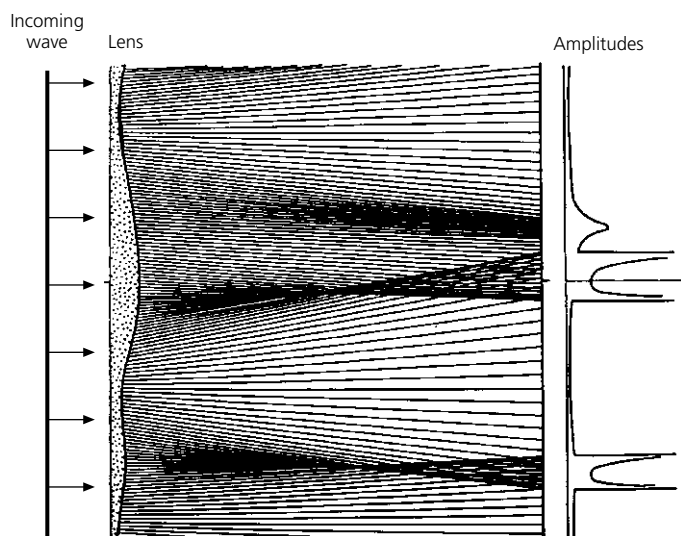


Fig. 3.7-5 An example of how velocity heterogeneities affect wave amplitudes. A plane wave impinging from the left is refracted by a layer of variable thickness. The amplitudes of the waves arriving at the right are shown. Regions of wide ray spacing have low amplitudes, and dense spacing yields large amplitudes. Concentrated lines, or caustics, cause very high amplitudes. (Hannay, 1986. Reproduced with permission from the Institute of Mathematics.)

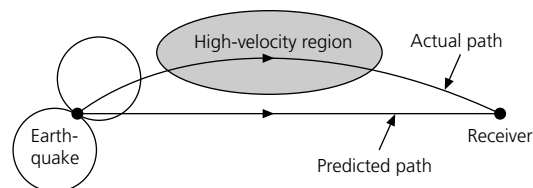


Fig. 3.7-6 Schematic example of how velocity heterogeneity can cause an erroneous estimate of either the focal mechanism or attenuation. The figure-eight structure at the earthquake shows the amplitude of a radiated surface wave as a function of azimuth, which depends on the focal mechanism (in this case dip-slip motion on a vertical fault). The predicted path would leave the source with a lower amplitude than the actual path, which is bent by the high-velocity region. Hence a focal mechanism study using these data without accounting for the perturbed ray path would be incorrect. Conversely, modeling the amplitudes without considering the high-velocity region would yield too-low estimates of attenuation.

The ray paths, which are normal to the local wave front, show how the initially planar wave is refracted. The ray spacing represents the energy density, so amplitudes are low where the rays are far apart, and high where they are close together. In some cases the energy focuses into caustics, areas of infinitely high energy density, which appear as solid black regions.

This example illustrates that velocity variations can affect the amplitudes of seismic waves some distance away. For example, small velocity heterogeneities near an earthquake can cause large amplitude variations at teleseismic distances. This

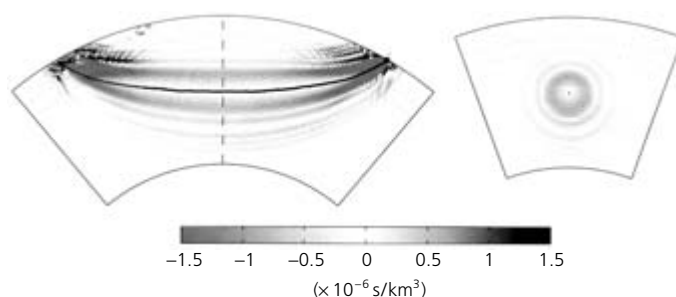


Fig. 3.7-7 Numerical simulation of the paths taken by seismic energy associated with the body wave phases S and sS for a 120 km-deep earthquake. The values shown, computed using normal modes, show the sensitivity of the travel time to velocity perturbations. These phases sample the structure in a banana-shaped region shown in side view (left) and end-on (right) surrounding the geometric ray path (solid line). (Zhao *et al.*, 2000.)

effect can be important, because most earthquakes occur at plate boundaries, such as subduction zones or mid-ocean ridges, where there are significant velocity heterogeneities. This phenomenon can cause difficulties in the interpretation of seismic data. For example, assume (Fig. 3.7-6) that the actual wave path from an earthquake to a receiver differs from that predicted due to a region of anomalously fast velocities. If the amplitudes of these waves were used to study the earthquake's focal mechanism, the result would be biased because the waves left the source in a direction different from that expected if the velocity heterogeneity were not present. Conversely, if the focal mechanism were known, the observed amplitude would differ from that expected, so an estimate of the attenuation would be incorrect.

When multipathing occurs, the seismic waves arriving at a receiver can be viewed as having taken some ray paths in addition to the direct path, and so have sampled a larger region of the earth. A way to view this is that Fermat's principle giving the geometric ray path applies exactly only to waves of infinite frequency. For waves of finite frequency, we can view the seismic waveform as a coherent sum of energy that travels all possible paths that arrive within a half-period of the infinite-frequency wave, which took the shortest time. These paths form a volume called the first *Fresnel zone* around the infinite-frequency path. Successive half-periods correspond to higher-order Fresnel zones. For longer-period waves, the maximum time over which energy arrives coherently is longer, so the Fresnel zones are proportionately larger. For example, teleseismic body waves sample a banana-shaped region about the geometric ray path. Figure 3.7-7 shows Fresnel zones for a body wave phase in a laterally homogeneous earth, plotted in terms of how the travel time is affected by velocity perturbations. The curved ray path represents the effects of vertical variations in velocity on the infinite frequency ray, and the surrounding "banana" represents the effects of finite-frequency waves. Lateral heterogeneity would distort the "banana."

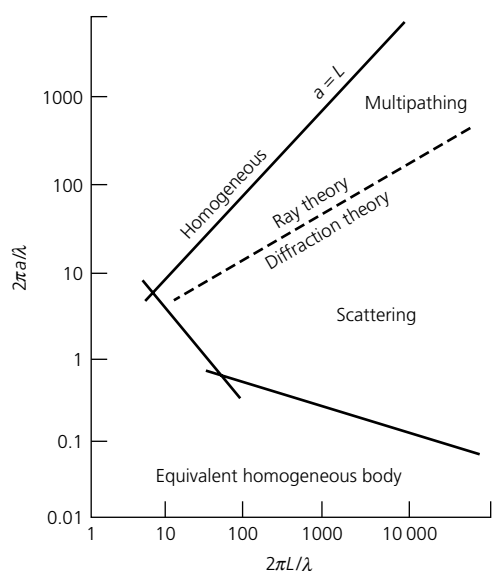


Fig. 3.7-8 Schematic representation of different approaches to seismic wave propagation in a medium with velocity heterogeneity. The approach depends on the ratio of the heterogeneity size a to the wavelength λ and the distance L the wave travels through the heterogeneous region. (After Aki and Richards, 1980. From *Quantitative Seismology*, © 1980 by W. H. Freeman and Company, used with permission.)

3.7.4 Scattering

A related effect to multipathing is the scattering of seismic waves. Both effects are complicated, and the distinction between them is gradational. As shown in Fig. 3.7-8, whether the effects of velocity heterogeneity are regarded as scattering depends on the ratio of the heterogeneity size to the wavelength and the distance the wave travels through the heterogeneous region. When the heterogeneity is large compared to the wavelength, we regard the wave as following a distinct ray path that is distorted by multipathing. When the velocity heterogeneities are closer in size to the wavelength, we think of scattered energy rather than distinct ray paths. However, when the heterogeneities are much smaller than the wavelength, they simply change the medium's overall properties. The further the wave travels in the heterogeneous region, the more useful the scattering description becomes. Hence for longer distances, the wavelength range viewed as scattering increases.³

Figure 3.7-8 also illustrates that diffraction can be viewed as behavior intermediate between scattering and multipathing.

³ The fact that light scattering in the atmosphere depends on wavelength and the distance traveled has familiar consequences. Because the shortest wavelengths of visible light are the most scattered, blue light reaching us from all directions makes the sky appear blue. The loss of blue light makes the sun appear yellow, although it would appear white if observed from a spacecraft. At sunset, when the sunlight passes through a longer path in the atmosphere than at other hours, intermediate wavelengths are also scattered, leaving direct light from the sun enhanced in the longest visible wavelengths (red light) and making the sun appear red.

As we have seen, some of the behavior of diffracted waves can be derived either using a Huygens' source scattering representation (Section 2.5.10) or by using ray paths in a medium with variable velocity, as for the head wave (Section 3.2.1) or core diffraction (Section 3.5.2). These ray paths were not truly geometric, in that energy was required to follow paths that did not obey Snell's law. The distinction between ray theory and diffraction depends on wavelength, as discussed in Section 2.5.10, so waves diffracted around the core are depleted in the higher frequencies.⁴

Scattering can be viewed in different ways. In some situations we view the scattering as deterministic, and try to image distinct scatterers. For example, migration methods in reflection seismology (Section 3.3.7) seek to undo the effects of scattering and produce a clearer image of the subsurface. In other situations, we view the medium as containing many scatterers and consider their effects on the wave field statistically. This approach is taken to the scattering of PKP waves (Fig. 3.5-8), with a wavelength of about 10 km, by lower mantle heterogeneities of about that size.

Scattering is especially important in the continental crust, which has many small layers and reflectors resulting from billions of years of continental evolution. Although these structures do not significantly affect waves with wavelengths longer than tens of km, for shorter-wavelength waves they can act as point scatterers or Huygens' sources. Hence some of the scattered energy arrives at a receiver after the initial pulse that obeyed Fermat's principle and took the shortest path. This scattered energy causes an arrival to have a *coda*, a tail of incoherent energy that decays over a duration of seconds or minutes. The main arrival has a polarity related to the direction of propagation that can be observed on a three-component seismometer by forming particle motion plots (Fig. 2.7-6). By contrast, the scattered energy arrives from various directions and thus shows little or no preferred particle motion.

Figure 3.7-9 demonstrates the scattering for a seismic arrival. The unscattered wave travels the shortest distance and gives the initial arrival (*left*). Scattered energy lost from this arrival that instead arrives later could have been scattered from an infinite number of locations that would yield the observed travel time. In a constant-velocity medium, the locus of these possible scatterers forms an ellipsoid with the source and the receiver as foci (*center*). Larger ellipsoids define the possible scatterers for energy that arrives later (*right*). These ellipsoids are distorted by velocity heterogeneity and are analogous to the Fresnel volume used when we consider the waves as following distinct ray paths.

Scattering is especially noticeable on the moon. Figure 3.7-10 contrasts seismic records of an earthquake and the impact of a rocket on the moon. Most of the earthquake's energy arrives in the main *P*- and *S*-wave arrivals. By contrast, on the moon

⁴ This effect makes it hard to understand what someone is saying when they are standing around a corner, because the voice sounds muffled due to the loss of the higher frequencies.

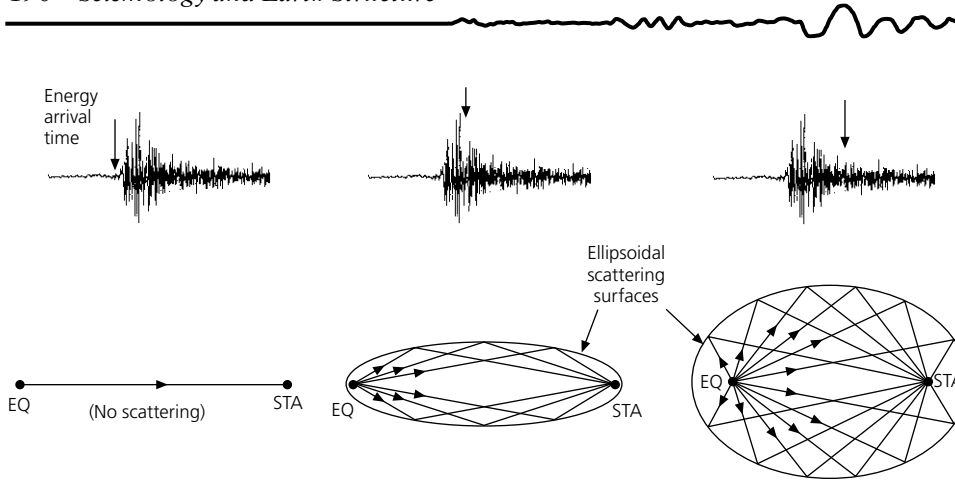


Fig. 3.7-9 Development of a P-wave coda due to scattering. *Left*: The first arrival follows the minimum-time path from the earthquake (EQ) to the station (STA) according to Fermat's principle, and involves no scattered energy. *Center*: Scattered energy arrives after the first arrival. An infinite number of possible locations for scatterers yield arrivals at this same time. In a homogeneous medium the locus of these points forms an ellipsoidal surface. *Right*: Energy arriving later in the coda can be modeled as arising from a larger ellipsoidal surface of possible scatterers.

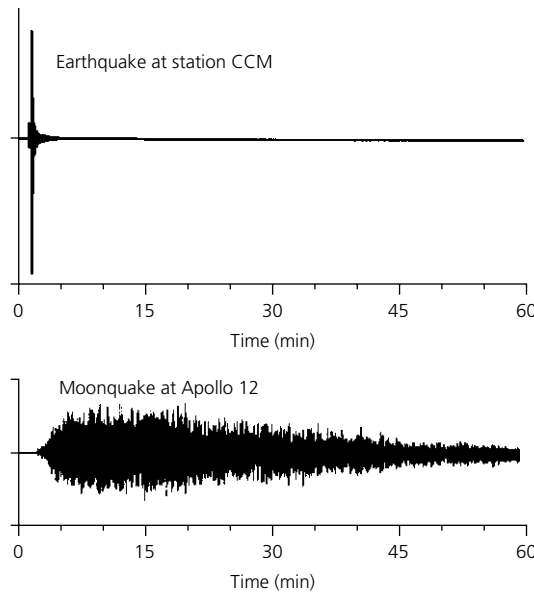


Fig. 3.7-10 Comparison of seismograms for the earth and the moon. *Top*: Seismogram recorded at Cathedral Cave, Missouri (CCM), from a small earthquake 183 km away. *Bottom*: Seismogram recorded by the Apollo 12 seismometer of the impact of the Apollo 14 Saturn booster rocket 147 km away. The terrestrial record shows high attenuation, whereas the lunar seismogram shows intense scattering due to the fractured regolith and very weak attenuation due to the lack of intergranular water. (Mitchell, 1995. *Rev. Geophys.*, 33, 441–62, copyright by the American Geophysical Union.)

the energy is intensely scattered, and no main arrivals can be identified. This is probably because intrinsic attenuation is much larger in the earth's crust than on the moon. The movements of interstitial fluids in the earth's crust greatly reduce seismic wave amplitudes, whereas energy scattered by the moon's highly fractured near-surface regolith layer is poorly absorbed and reverberates. As a result, efforts to identify seismic phases and use them to study the moon's internal structure have been generally unsuccessful.

3.7.5 Intrinsic attenuation

We can gain insight into the intrinsic attenuation of seismic waves by examining a simple system, a damped harmonic oscillator composed of a spring and a dashpot. We use Newton's second law, $F = ma$, to describe the displacement $u(t)$ of a mass m . The restoring force of the spring is proportional to minus the spring constant k times the spring extension or displacement from the equilibrium positions, so

$$m \frac{d^2 u(t)}{dt^2} + k u(t) = 0. \quad (2)$$

Once set in motion by an impulse, this frictionless system has a purely elastic response described by a perpetual harmonic oscillation

$$u(t) = A e^{i\omega_0 t} + B e^{-i\omega_0 t}, \quad (3)$$

where A and B are constants, and the mass moves back and forth with a natural frequency

$$\omega_0 = (k/m)^{1/2}. \quad (4)$$

One example of this general solution is

$$u(t) = A_0 \cos(\omega_0 t). \quad (5)$$

Once the motion is started, this undamped oscillation continues forever, because no energy is lost. However, this is no longer the case if the system contains a dashpot, or damping term. The damping force is proportional to the velocity of the mass and opposes its motion. Hence the equation of motion (Eqn 2) becomes

$$m \frac{d^2 u(t)}{dt^2} + \gamma m \frac{du(t)}{dt} + k u(t) = 0, \quad (6)$$

where γ is the *damping factor*. To simplify this, we define the *quality factor*

$$Q = \omega_0/\gamma, \quad (7)$$

and rewrite Eqn 6 as

$$\frac{d^2 u(t)}{dt^2} + \frac{\omega_0}{Q} \frac{du(t)}{dt} + \omega_0^2 u(t) = 0. \quad (8)$$

This differential equation, which describes the damped harmonic oscillator, can be solved assuming that the displacement is the real part of a complex exponential

$$u(t) = A_0 e^{ipt}, \quad (9)$$

where p is a complex number. Substituting Eqn 9 into Eqn 8 yields

$$(-p^2 + ip\omega_0/Q + \omega_0^2) A_0 e^{i(pt)} = 0. \quad (10)$$

For this to be satisfied for all values of t ,

$$-p^2 + ip\omega_0/Q + \omega_0^2 = 0. \quad (11)$$

Because p is complex, we break it into its real and imaginary parts,

$$p = a + ib, \quad p^2 = a^2 + 2iab - b^2, \quad (12)$$

so Eqn 11 gives

$$-a^2 - 2iab + b^2 + ia\omega_0/Q - b\omega_0/Q + \omega_0^2 = 0, \quad (13)$$

which can be split into equations for the real and imaginary parts and solved separately:

$$\text{Real:} \quad -a^2 + b^2 - b\omega_0/Q + \omega_0^2 = 0, \quad (14)$$

$$\text{Imaginary:} \quad -2ab + a\omega_0/Q = 0.$$

Solving the imaginary part for b gives

$$b = \omega_0/2Q, \quad (15)$$

and putting this into the equation for the real part gives

$$a^2 = \omega_0^2 - \omega_0^2/4Q^2 = \omega_0^2(1 - 1/4Q^2). \quad (16)$$

Thus we define

$$\omega = a = \omega_0(1 - 1/4Q^2)^{1/2}, \quad (17)$$

and rewrite Eqn 9 with separate real and imaginary parts,

$$u(t) = A_0 e^{i(\omega t + ibt)} = A_0 e^{-bt} e^{i\omega t}. \quad (18)$$

The real part is the solution for the damped harmonic displacement,

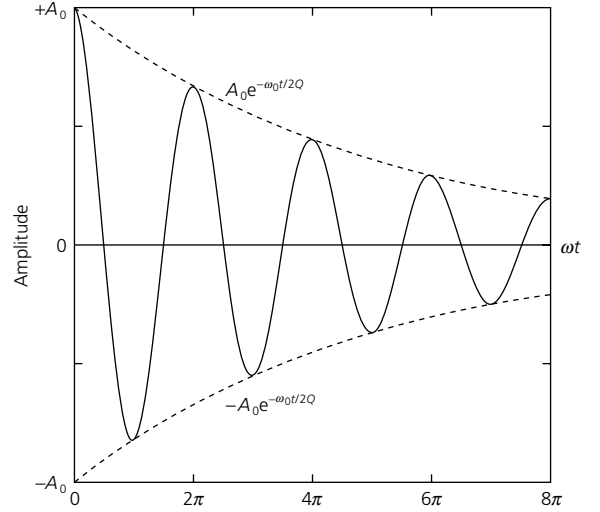


Fig. 3.7-11 For a damped harmonic oscillator, the envelope (dashed lines) amplitude is initially A_0 , but decays with time at a rate determined by the quality factor, Q .

$$u(t) = A_0 e^{-\omega_0 t/2Q} \cos(\omega t). \quad (19)$$

This solution shows how the damped oscillator responds to an impulse at time zero (Fig. 3.7-11). It is no longer a simple harmonic oscillation because it differs in two ways from the undamped solution (Eqn 5). The exponential term expresses the decay of the signal's envelope, or overall amplitude,

$$A(t) = A_0 e^{-\omega_0 t/2Q}, \quad (20)$$

which is superimposed on the harmonic oscillation given by the cosine term. Moreover, the frequency of the harmonic oscillation (Eqn 17) is changed from the natural frequency of the undamped system, ω_0 , by an amount depending on the quality factor. Q is inversely proportional to the damping factor, γ , so the smaller the damping, the greater Q is. For no damping, Q is infinite, and the damped solution reduces to the undamped one, because its amplitude does not decay with time (Eqn 20), and its frequency remains ω_0 (Eqn 17). As the damping increases, Q decreases, so the amplitude decays faster, and the frequency changes more from its undamped value. Equation 20 shows that the amplitude decays to e^{-1} (0.37) of its original value by the *relaxation time*

$$t_{1/e} = 2Q/\omega_0. \quad (21)$$

Because the energy in an oscillating system is proportional to the square of the amplitude, as we saw for a harmonic wave in Section 2.2.4, Eqn 20 gives the energy of the oscillator as

$$E(t) = \frac{1}{2} k A^2(t) = \frac{1}{2} k A_0^2 e^{-\omega_0 t/Q} = E_0 e^{-\omega_0 t/Q}. \quad (22)$$

Energy decays faster than the amplitude, because the negative exponent in Eqn 22 is twice as large as in Eqn 21.

3.7.6 Quality factor, Q

The solution for the damped harmonic oscillator incorporated the damping through the quality factor, Q . Attenuation for seismic waves and a variety of other physical phenomena are often discussed in terms of Q or Q^{-1} . Although Q has more convenient values, Q^{-1} has the advantage that it is directly, rather than inversely, proportional to the damping. In some cases, Q is used to describe the decay of an oscillation, whereas in others it is used to describe the physical properties of the system that cause a disturbance to attenuate. For example, the Q of one of the earth's normal modes, which is directly analogous to a damped oscillator, describes how the mode decays with time. This decay results from the distribution of material in the earth that causes seismic energy to be lost to heat. This distribution can be described in terms of a Q , or anelastic attenuation, structure analogous to the elastic velocity structure.

As a result, we speak of the Q of surface waves, body waves, and crustal phases like Lg . We also speak of the variation within the earth of Q_α and Q_β , which controls the attenuation of P and S waves. The anelastic structure of the earth, given by variations in Q_α and Q_β , is analogous to the elastic velocity structure because Q can be viewed mathematically as an imaginary part of the velocity. To see this, note that (9), which we used to derive the decaying oscillation, can be viewed as an oscillation with a complex frequency p

$$u(t) = A_0 e^{ipt} = A_0 e^{i(a+ib)t} \quad (23)$$

where the real and imaginary parts of the frequency are

$$a = \omega \quad b = \omega^* = \omega_0/2Q \approx \omega/2Q \quad (24)$$

assuming that attenuation is small (Q large) enough that $\omega \approx \omega_0$. Hence we write

$$Q^{-1} = 2b/a = 2\omega^*/\omega. \quad (25)$$

Treating the attenuation as an imaginary part of the frequency and dividing by the wavenumber lets us treat the corresponding velocity for a propagating wave as complex,

$$c + ic^* = \omega/k + i\omega^*/k = \omega/k + i\omega Q^{-1}/2k \quad (26)$$

so

$$Q^{-1} = 2c^*/c. \quad (27)$$

Thus we can express the attenuation of P - and S -waves by using the quality factors Q_α and Q_β to give imaginary parts to the velocities. If there is no attenuation ($Q = \infty$) the frequency and

the velocity have no imaginary parts. This formulation is useful because it means that methods used to invert surface wave velocities or normal mode eigenfrequencies to find velocity in the earth can also be used to invert observations of their attenuation to find the distribution of anelasticity.

We pose the complex parts of the velocities in terms of the properties of the material causing attenuation by treating the elastic moduli as having imaginary parts. For the shear velocity

$$\begin{aligned} \beta + i\beta^* &= \beta(1 + iQ_\beta^{-1}/2) \\ &= ((\mu + i\mu^*)/\rho)^{1/2} = \beta(1 + i\mu^*/\mu)^{1/2} \\ &\approx \beta(1 + i\mu^*/2\mu) \end{aligned} \quad (28)$$

where the last step used the first term of the Taylor series, because the attenuation and hence imaginary part is small. Comparing terms shows that

$$Q_\beta^{-1} = \mu^*/\mu. \quad (29)$$

A similar analysis shows that the quality factor for P waves is given by the imaginary parts of both the bulk and shear moduli

$$Q_\alpha^{-1} = (K^* + 4/3\mu^*)/(K + 4/3\mu). \quad (30)$$

Physically, it is useful to think of energy as being lost in either compressional or shear deformation, so we express their attenuation in terms of imaginary parts of the compressibility and rigidity

$$Q_K^{-1} = K^*/K \quad Q_\mu^{-1} = \mu^*/\mu = Q_\beta^{-1}. \quad (31)$$

These quality factors are related to those for the velocities by

$$Q_\alpha^{-1} = LQ_\mu^{-1} + (1-L)Q_K^{-1} \quad L = (4/3)(\beta/\alpha)^2. \quad (32)$$

In general little energy is lost in compression, so Q_K^{-1} is very small, and thus most of the attenuation for P waves occurs in shear, making $Q_\alpha^{-1} \approx (4/9)Q_\beta^{-1}$.

Techniques for measuring Q in the earth follow from those used to measure Q for the decay of an oscillation. From Eqn 20, taking the natural logarithm of the envelope shows that

$$\ln A(t) = \ln A_0 - \omega_0 t/2Q, \quad (33)$$

so Q can be found from the slope of the logarithmic decay. Alternatively, if successive peaks one full period $T = 2\pi/\omega_0$ apart have amplitudes

$$\begin{aligned} A_1(t_1) &= A_0 \exp(-\omega_0 t_1/2Q) \quad \text{and} \\ A_2(t_1 + T) &= A_0 \exp(-\omega_0(t_1 + T)/2Q), \end{aligned} \quad (34)$$

their ratio is

$$A_1/A_2 = \exp[-\omega_0 t_1/2Q - \omega_0(t_1 + T)/2Q] = \exp(\pi/Q), \quad (35)$$

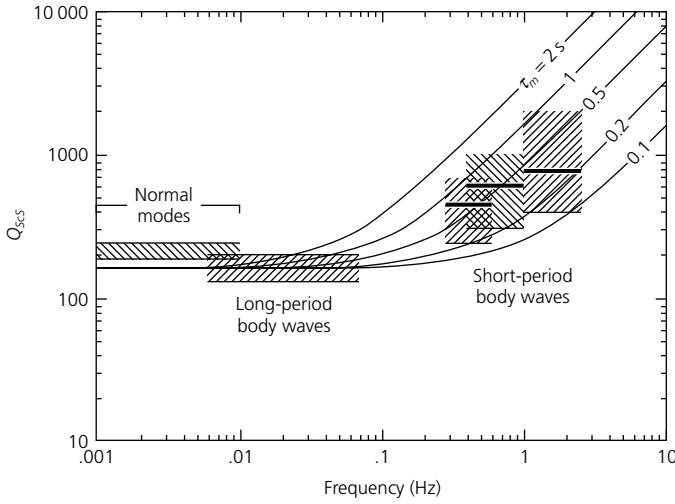


Fig. 3.7-12 Frequency dependence of attenuation for seismic waves in the mantle. Q is shown as though all measurements were for ScS waves, a good measure of the average mantle value because of their path from surface to core and back. (Sipkin and Jordan, 1979. © Seismological Society of America. All rights reserved.)

so

$$Q = \pi / \ln (A_1/A_2). \quad (36)$$

To illustrate this approach, note that in Fig. 3.7-11 the second peak, at $\omega t = 2\pi$, is about 2/3 of the first peak, at $\omega t = 0$. Thus $Q \approx \pi / \ln (3/2) \approx 8$. This is small compared to Q for mantle rocks, which is in the range of 200–500, but comparable to that for some sedimentary rocks. For example, S waves in shale have $Q \approx 10$.

Another way to view Q is as the number of cycles the oscillation takes to decay to a certain level. The number of cycles n , is

$$n = t/T = \omega t / 2\pi \approx \omega_0 t / 2\pi, \quad (37)$$

where the last approximation, based on Eqn 17, assumes that the attenuation is small enough ($Q \gg 1$) so that $\omega \approx \omega_0$. The amplitude at time t_n , after n cycles, is

$$A(t_n) \approx A_0 e^{-\frac{n\pi}{Q}}, \quad (38)$$

so, if we define n as equal to Q ,

$$A(t_n) \approx A_0 e^{-\pi} \approx 0.04 A_0. \quad (39)$$

Thus, after Q cycles, the amplitude drops to a level of $e^{-\pi}$ or 4% of the original amplitude. Hence, in Fig. 3.7-11, more than 95% of the amplitude is lost after $Q \approx 8$ cycles.

Q can describe the oscillation's decay in either time or space. For standing waves like normal mode oscillations, Q describes the decay of amplitudes with time. For traveling waves, we

replace t with x/c , where x is the distance traveled and c is the velocity. Thus Eqn 20 becomes

$$A(x) = A_0 e^{\frac{-\omega_0 x}{2cQ}}, \quad (40)$$

which describes how the amplitude decays with the distance the wave propagates.

When these techniques are used to measure Q for seismic waves, we find that Q varies with frequency (Fig. 3.7-12). Q is essentially constant at low frequencies, about 0.001 to 0.1 Hz, but then increases with frequency. Thus Q values derived from normal mode analysis are lower than those obtained from higher-frequency waves. Although our first instinct might be that Q should be frequency-independent, such a situation imposes a stringent requirement. Because $Q = \omega/\gamma$, constant Q requires a physical mechanism in the earth with damping proportional to frequency. We will explore this issue shortly.

Before doing so, it is also worth noting that our model of the damped oscillator assumes that the attenuation is *linear*, such that Q is independent of the amplitude of the wave. This is the same as assuming that the amplitudes are not too large. In most rocks this condition is satisfied if the strains involved with the wave propagation are less than about 10^{-6} . Although this is true at teleseismic distances, it is not the case near an earthquake or an explosion, where the elastic strain can exceed 10^{-4} . Large earthquakes can cause large strains, and hence a region of nonlinear attenuation.

3.7.7 Spectral resonance peaks

We are interested in understanding how anelasticity in the earth causes the attenuation of propagating waves. This behavior is an example of the general case of how a damped harmonic oscillator responds to a driving force that depends on frequency. To see this, we modify Eqn 8 by adding a harmonic driving force, and so have the inhomogeneous equation

$$\frac{d^2 u}{dt^2} + \gamma \frac{du}{dt} + \omega_0^2 u = e^{i\omega t}. \quad (41)$$

The solution is found using a trial solution

$$u(t) = A(\omega) e^{i\phi(\omega)} e^{i\omega t}. \quad (42)$$

Substituting this in Eqn 41 yields the amplitude response, $A(\omega)$, and phase response, $\phi(\omega)$,

$$A(\omega) = [(\omega_0^2 - \omega^2)^2 + (\omega\gamma)^2]^{-1/2}, \quad \phi(\omega) = \tan^{-1} \left[\frac{-\gamma\omega}{\omega_0^2 - \omega^2} \right]. \quad (43)$$

As shown in Fig. 3.7-13, the amplitude and the phase responses depend on the damping factor γ and how far the forc-

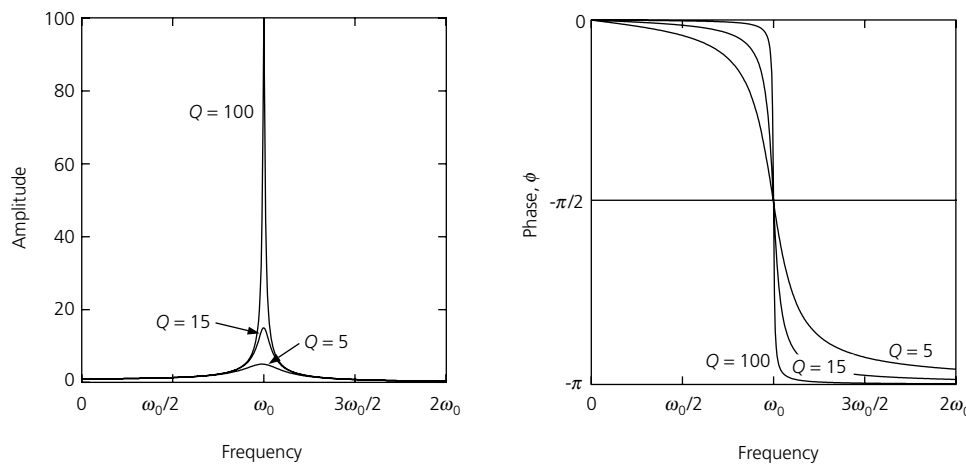


Fig. 3.7-13 Amplitude (*left*) and phase (*right*) response of a forced, damped harmonic oscillator with natural frequency ω_0 . For greater damping (lower Q) the peak decreases and both it and the phase shift are broadened from the sharp values they have with little damping.

ing frequency ω is from the oscillator's natural or *resonant* frequency, ω_0 . The resonance curve shows how the damped harmonic oscillator responds to the frequency-dependent driving force. The closer the driving frequency ω is to the oscillator's natural frequency ω_0 , the more the oscillator responds.

The resonance curve can be viewed in terms of the frequency at which the peak occurs

$$\omega_p = (\omega_0^2 - \gamma^2/2)^{1/2} = \omega_0(1 - 1/2Q^2)^{1/2} \quad (44)$$

and the amplitude of the peak

$$A(\omega_p) = Q/(\omega_0^2(1 - 1/4Q^2)^{1/2}). \quad (45)$$

If the oscillator is undamped ($\gamma = 0$, $Q = \infty$) the peak occurs at its natural frequency and shows an infinite response. Adding damping lowers the amplitude of the peak and shifts it. However, the shift is very small unless the system is much more damped ($Q < 2$) than occurs for seismic wave attenuation. The damping also spreads out the peak in frequency, so the more the damping, the broader and lower the peak. To see why, recall that the more the damping, the faster the oscillation decays as a function of time (Fig. 3.7-11). As we will see in Chapter 6, the spectrum of an undamped sinusoid is a sharp line, or delta function, so additional frequencies, and thus a broader peak, correspond to the decaying sinusoid. The phase response also has significance, as we will see when we discuss seismometers (Section 6.6).

The resonance curve concept appears in a wide variety of applications, because many physical systems can be viewed as damped harmonic oscillators. Three commonly considered in seismology are the attenuation of the earth's normal modes, the behavior of a seismometer, and the response of a building to ground motion. An earthquake puts energy at various frequencies into the earth, exciting its normal modes (Section 2.9). These modes form a set of damped harmonic oscillators, so the amplitude spectrum of a long-period seismogram contains

peaks that correspond to the net resonance curve for each mode multiplet. The width of a peak depends on the frequencies and amplitudes of the mode's singlets and the mode's damping. Seismometers can also be viewed as damped harmonic oscillators, whose natural frequency and damping control their response to ground motion. In addition, as mentioned in Section 1.2.2, buildings can be considered damped harmonic oscillators. This concept is important in designing earthquake-resistant structures, because buildings are most vulnerable to ground motion with frequencies close to their natural frequencies, so damping is added to reduce the resulting motion.

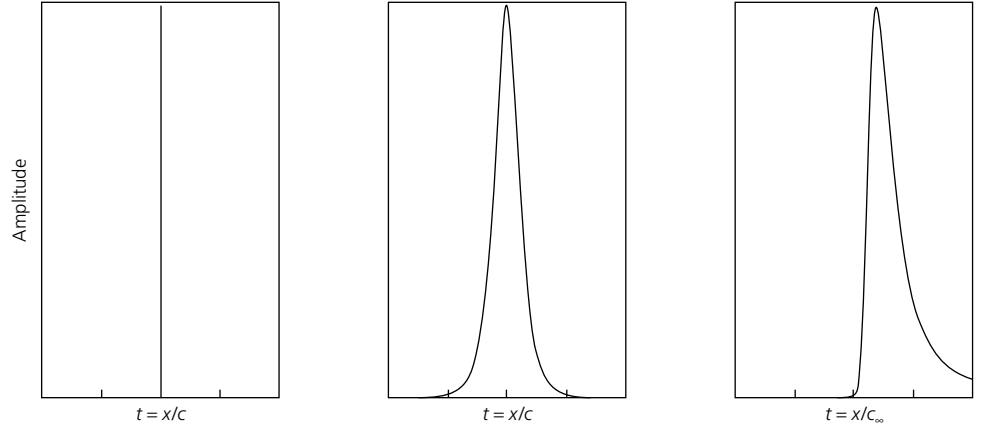
3.7.8 Physical dispersion due to anelasticity

An important consequence of seismic wave attenuation is *physical dispersion*, in which waves at different frequencies travel at different velocities. This differs from the *geometrical dispersion* discussed in Sections 2.7 and 2.8, in which surface waves of different frequencies have different apparent velocities at the surface because they sample different depths and hence encounter material of different velocities. Thus, although the intrinsic velocity of the rock at any depth is treated as frequency-independent, dispersion occurs because of the depth-variable velocity of the material. By contrast, with physical dispersion the intrinsic velocity of waves in the medium varies with frequency.⁵

To see how physical dispersion results from attenuation, consider how a seismic wave changes shape. Assume that a delta function wave, a pulse of infinite height and unit area (Fig. 3.7-14), propagates through a homogeneous elastic medium with intrinsic velocity c :

⁵ The rainbow results from physical dispersion for light waves passing through water drops in the atmosphere or a prism. Different frequencies (colors) of light travel at different speeds through the water or prism, and thus refract at different angles, separating initially white light into different colors.

Fig. 3.7-14 *Left*: A propagating wave pulse composed of a delta function. With no dispersion, all frequencies arrive at the same time. *Center*: The delta function after broadening by attenuation, showing that energy arrives before the high-frequency arrival time. *Right*: The pulse including physical dispersion, which makes the lower-frequency waves travel more slowly, so that they do not arrive before the highest-frequency component.



$$u(x, t) = \delta(t - x/c). \quad (46)$$

The Fourier transform of the delta function,

$$F(\omega) = \int_{-\infty}^{\infty} u(x, t) e^{-i\omega t} dt = \int_{-\infty}^{\infty} \delta(t - x/c) e^{-i\omega t} dt = e^{-i\omega x/c}, \quad (47)$$

shows that the delta function is made up of waves of all frequencies, as we discuss further in Section 6.2.5. If there is no dispersion, all the frequencies travel at the same speed and arrive at the same time. The effect of attenuation as a function of distance is given by writing Eqn 40 as a function of frequency,

$$A(\omega) = e^{\frac{-\omega x}{2cQ}}, \quad (48)$$

which shows that if Q is constant, the rate at which the amplitude decays with distance increases strongly with frequency. To see how this attenuation affects the delta function wave, we multiply Eqn 47 by Eqn 48 and use the inverse Fourier transform to return to the time domain

$$u(x, t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} A(\omega) F(\omega) e^{i\omega t} d\omega = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{\frac{-\omega x}{2cQ}} e^{\frac{-i\omega x}{c}} e^{i\omega t} d\omega. \quad (49)$$

Evaluating the integral yields

$$u(x, t) = [(x/2cQ)/((x/2cQ)^2 + (x/c - t)^2)]/\pi, \quad (50)$$

so the delta function is broadened by attenuation into a wavelet that is symmetric in time about its maximum at $t = x/c$ (Fig. 3.7-14, *center*).

A problem with this solution is that seismic energy arrives before the geometric arrival time of the delta function pulse,

$t = x/c$, which is the arrival time of the infinite-frequency component. In fact, because the tails of the wavelet extend to infinity on both sides of $t = x/c$, some energy arrives before the earthquake occurred. This impossible situation, called *noncausality*, results from the fact that attenuation broadened the pulse by preferentially removing the high-frequency components.⁶

Thus the physical mechanisms that cause attenuation in the earth must prevent waves of all frequencies from traveling at the same speed. Instead, there must be dispersion, where the lower frequencies causing the tails travel more slowly and arrive later. We saw in Section 2.8 that in a dispersive medium we distinguish the phase velocity c , the speed of a wave of a single frequency, from the group velocity that describes the speed of a wave group. Thus the mathematical condition for causality is that $u(x, t) = 0$ for all $t < x/c_\infty$, where $c_\infty = c(\infty)$ is the phase velocity of the infinite-frequency waves that arrive first. One such dispersion relation for phase velocity as a function of frequency, called Azimi's attenuation law, is

$$c(\omega) = c_0 \left[1 + \frac{1}{\pi Q} \ln \left(\frac{\omega}{\omega_0} \right) \right], \quad (51)$$

where c_0 is a reference velocity corresponding to a reference frequency ω_0 .⁷ This relation provides the needed causality, because the resulting pulse (Fig. 3.7-14, *right*) has high frequencies arriving at or soon after $t = x/c_\infty$, whereas the low frequencies arrive later over a duration depending on the value of Q . If there is no attenuation ($Q = \infty$), Eqn 51 yields no dispersion, and the delta function is not broadened.

From Eqn 51, the P - and S -wave velocities α and β vary as a function of period T , as

⁶ We noted a similar effect in Section 2.9.8: namely, that individual normal modes of a single frequency appear to predict displacement before a wave could arrive, but their sum gives a wave at the correct time.

⁷ Aki and Richards (1980).

$$\beta(T) = \beta(1) \left(1 - \frac{\ln T}{\pi} Q_\mu^{-1} \right),$$

$$\alpha(T) = \alpha(1) \left[1 - \frac{\ln T}{\pi} \left(L Q_\mu^{-1} + (1-L) Q_K^{-1} \right) \right],$$

$$\text{where } L = \frac{4}{3} \left(\frac{\beta}{\alpha} \right)^2, \quad (52)$$

where $\alpha(1)$ and $\beta(1)$ are the velocities at 1 s. We find how a wave's travel time varies with period by integrating along its ray path (Eqn 3.4.16). The effect can be significant. For a vertical *ScS* wave, the travel time for $T = 40$ s is 5 s slower than for $T = 1$ s. Out of a total travel time of about 934 s, this is a difference of 0.5%. For vertical *PcP* waves at the same periods, the travel time difference is 1 s out of 511 s, or 0.2%.

This phenomenon causes a discrepancy between the seismic velocity structure found by inverting observations of long-period normal modes and short-period body waves. The velocities inferred from normal modes are consistently slower than those from body waves. The discrepancy reflects the fact that attenuation causes longer-period waves that are studied as normal modes to travel at lower velocities than the body waves. Failure to take this effect into account can cause errors in the predicted arrival times of body waves of several seconds.

The pulse in Fig. 3.7-14 (*right*) is also known as an *attenuation operator*, and can be used to model the effects of attenuation on seismic waveforms. As discussed in Section 3.3.6 and derived in Section 6.3, seismic signals can be modeled by convolving the source-time function with operators describing different effects. Thus a synthetic seismogram computed for an elastic earth can be convolved with the attenuation operator to create a more realistic pulse.

Body wave attenuation is often characterized using the parameter t^* . If a ray travels through a region of constant Q ,

$$t^* = \frac{t}{Q} = \frac{\text{travel time}}{\text{quality factor}}. \quad (53)$$

Because Q varies within the earth, we derive t^* by integrating along the ray path,

$$t^* = \int \frac{dt}{Q} = \sum_{i=1}^N \frac{\Delta t_i}{Q_i}, \quad (54)$$

where Δt_i and Q_i are the travel time and Q values on the i^{th} path segment. For *P* waves, t_α^* is often about 1 s, whereas *S* waves typically have t_β^* around 4 s. The values of t^* increase with increased distance, but are also affected by the number of passages through the asthenosphere (about 80–220 km depth). For example, *ScS* tends to have a higher t^* (greater attenuation) than *S* at the same distance because of the longer ray path, and

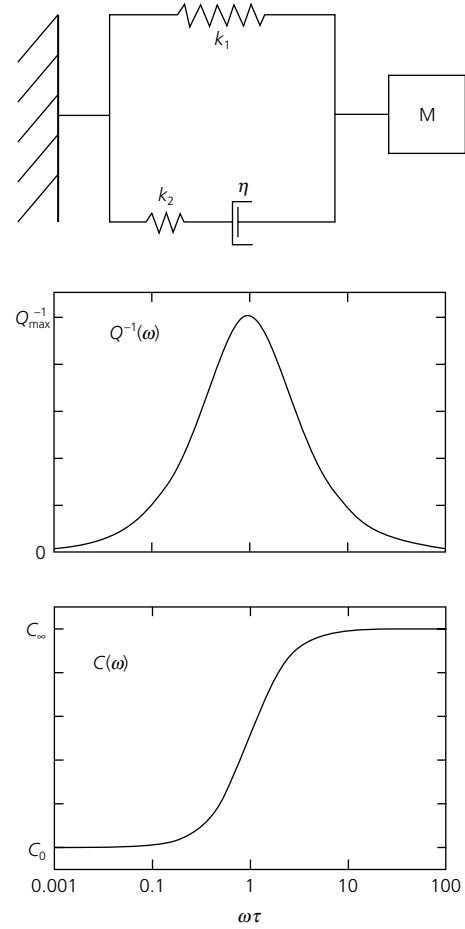


Fig. 3.7-15 *Top*: Schematic diagram of a standard linear solid, made up of a mass connected to two springs and a dashpot. This system responds elastically to waves with periods that are short compared to the relaxation time, τ , and viscously for periods longer than the relaxation time. *Center*: Absorption peak for this material. Q^{-1} approaches zero for large and small values of $\omega\tau$ and is greatest for $\omega\tau = 1$. *Bottom*: Phase velocity dispersion resulting from the attenuation. The velocity is c_0 at low frequencies and increases to c_∞ at high frequencies.

S waves from deep earthquakes that only cross the asthenosphere once have lower t^* than *S* waves from shallow events.

3.7.9 Physical models for anelasticity

A common model for the anelastic processes in the earth causing attenuation treats the material as a *viscoelastic* or *standard linear solid*, which combines elastic and viscous responses to an incident seismic wave. This model is represented by a spring with constant k_1 in parallel with a spring with constant k_2 and a dashpot with viscosity η (Fig. 3.7-15). If a step function strain $H(t)$ (0 for $t < 0$, 1 afterwards) is applied, the stress response includes an instantaneous elastic contribution from spring k_1 and a delayed response from the dashpot and spring k_2 ,

$$\sigma(t) = k_1 H(t) + k_2 e^{-t/\tau}, \quad (55)$$

where τ is the relaxation time constant $= \eta/k_2$.

The response to harmonic waves depends on the product of the angular frequency and the relaxation time. For wave periods that are very short compared to the relaxation time, the system responds mostly elastically, and there is little attenuation. For wave periods much longer than the relaxation time, the system responds mostly in a viscous manner, so there is no attenuative loss of energy. As shown in Fig. 3.7-15, the attenuation⁸ varies as

$$Q^{-1}(\omega) = \frac{k_2}{k_1} \frac{\omega\tau}{1 + (\omega\tau)^2}. \quad (56)$$

At very low or very high frequencies $Q^{-1}(\omega)$ approaches zero, so Q becomes infinite. The greatest attenuation, or *absorption peak*,⁹ occurs at $\omega\tau = 1$, where

$$Q_{\max}^{-1} = Q^{-1}(1/\tau) = k_2/2k_1. \quad (57)$$

The phase velocity also depends on $\omega\tau$:

$$c(\omega) = c_0 \left[1 + \frac{k_2}{2k_1} \frac{(\omega\tau)^2}{1 + (\omega\tau)^2} \right], \quad (58)$$

where $c_0 = (k_1/\rho)^{1/2}$. The phase velocity is lowest (c_0) at low frequencies, and reaches

$$c_\infty = c_0(1 + k_2/2k_1) = c_0(1 + Q_{\max}^{-1}) \quad (59)$$

at high frequencies. This model thus has the key feature of the physical dispersion relation (Eqn 51) discussed earlier, that long-period waves travel more slowly than high-frequency waves.

Given this model, the fact that seismological observations find relatively constant Q over a large range of low frequencies from about 0.001 to 0.1 Hz (Fig. 3.7-12) is surprising. Moreover, theoretical and laboratory studies of the physical mechanisms thought to cause attenuation in the earth also suggest that Q should be strongly frequency-dependent. Hence the relatively constant value at low frequencies is thought to result from the superposition of many different mechanisms. A possible explanation comes from noting that a typical attenuation spectrum for a polycrystalline structure (Fig. 3.7-16, *top*) contains multiple attenuation peaks or absorption bands. The absorption bands depend on the material's composition and grain size and vary with temperature (recall Fig. 3.7-2) and pressure, such that higher pressure decreases attenuation, whereas

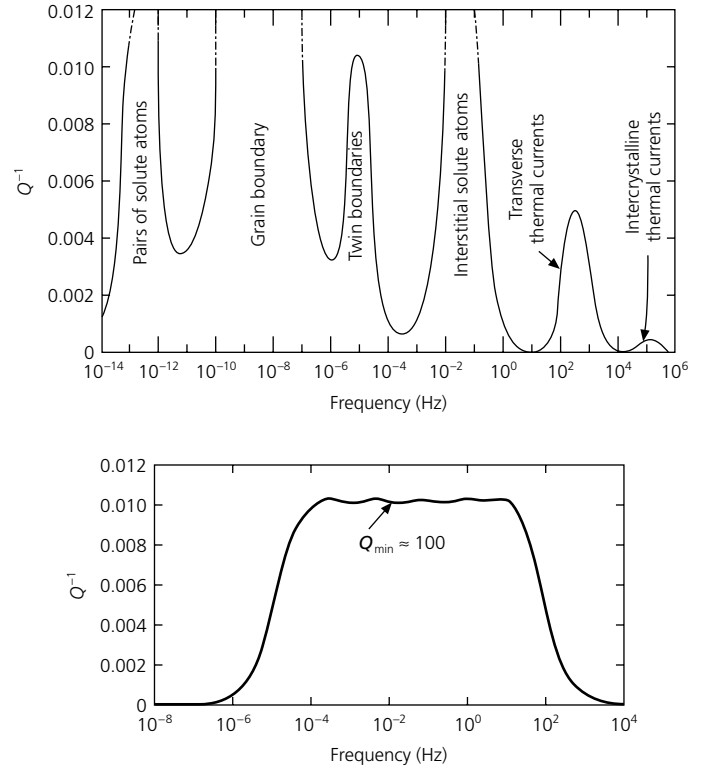


Fig. 3.7-16 *Top:* Relaxation spectrum for a polycrystalline material showing attenuation peaks at different frequencies due to different microscopic mechanisms. *Bottom:* Schematic model to explain the observation that Q is roughly constant over a wide range of frequencies. The superposition of absorption peaks for different compositions at different temperatures and pressures yields a flat absorption band. (Liu *et al.*, 1976.)

higher temperature increases it. Waves of various frequencies traversing the earth may feel the net effect of absorption bands with different relaxation times, yielding a flat absorption spectrum (Fig. 3.7-16, *bottom*). The higher-frequency waves in Fig. 3.7-12 that show a frequency-dependent Q would be above the flat part of the absorption spectrum.

3.7.10 Q from crust to inner core

Attenuation is inferred in all regions of the earth except for the liquid iron outer core, and varies greatly both laterally and vertically. In the crust, the greatest attenuation (lowest Q or highest Q^{-1}) occurs near the surface (Fig. 3.7-17), presumably due to the presence of fluids. Attenuation is lowest at about 20–25 km depth, and then increases again, presumably due to increasing temperature. Attenuation decreases as a function of frequency, as in Fig. 3.7-12, and varies geographically. Q in the upper crust is roughly proportional to the time since the last major tectonic activity in a region, perhaps due to crack generation and fluid flow during tectonism and gradual crack annealing after tectonism ceased.

⁸ Kanamori and Anderson (1977).

⁹ This effect is like driving over a bump: at a high speed inertia keeps the car in line and the bump is not very noticeable. At low speed, we feel only a gradual swell in the road. However, at an intermediate speed the bump gives the maximum jolt.

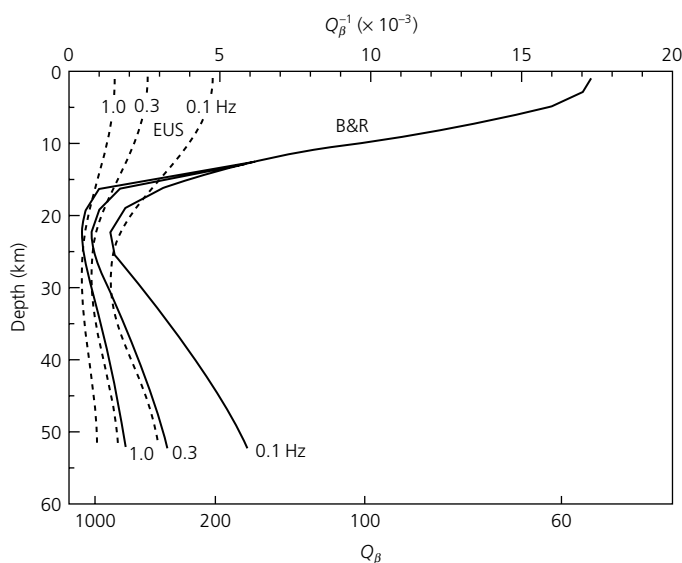


Fig. 3.7-17 Variation in attenuation with lithospheric depth for the eastern USA (EUS) and Basin and Range (B&R). Lower attenuation occurs for higher frequencies. (Mitchell, 1995. *Rev. Geophys.*, 33, 441–62, copyright by the American Geophysical Union.)

Regional variations in crustal Q are often studied with L_g waves, a superposition of higher-mode surface waves that give prominent arrivals in continental regions. Q_{L_g} for the USA varies regionally (Fig. 3.7-18), with values as high as 750 in the stable East and as low as 250 in the tectonically active West. This regional difference in attenuation, also seen in Figs 3.7-1 and 3.7-17, has implications for seismic hazards (Section 1.2.2). Similarly, the fact that the USA tested nuclear weapons in the western USA, which is more attenuative than the areas used by the Soviet Union, is significant for verifying test ban treaties (Section 1.2.8).

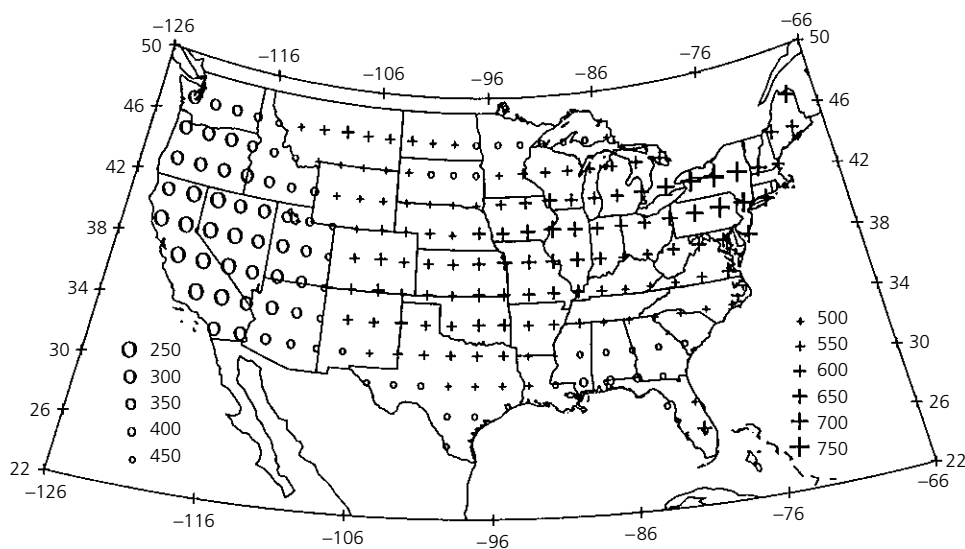


Fig. 3.7-18 Q_{L_g} for the USA mapped from the codas of 1 Hz L_g waves. Q_{L_g} , which reflects attenuation within the crust, shows higher attenuation in the tectonically active western USA and lower attenuation in the tectonically inactive east. (Mitchell *et al.*, 1997.)

Attenuation in the upper mantle varies with depth, with the lowest Q in the asthenosphere from about 80 to 220 km depth (Fig. 3.7-19). At these depths the temperature approaches, and perhaps exceeds, the melting temperatures of rock, so a small percentage of partial melt may exist. This pattern of attenuation is similar to that for seismic velocities, which are lowest in the asthenosphere. Hence both the elastic velocity and anelastic attenuation reflect the physical processes causing the mechanically weak asthenosphere. Beneath the asthenosphere, Q increases gradually with depth, presumably because temperature increases at a slower rate than pressure.

Q_μ increases with depth through the lower mantle, reaching values in excess of 500. There is some indication that attenuation is enhanced in the D'' region at the base of the mantle. Although no attenuation of P waves is detected for the outer core, there is significant attenuation of $PKIKP$ waves traversing the inner core, yielding Q_K estimates in the range of 150–300.

Lateral variations in attenuation are studied using tomographic methods similar to those used for velocity (Sections 2.8.3, 7.3). Where temperatures vary over short distances, significant attenuation variations can occur, as shown in Fig. 3.7-3 for a mid-ocean ridge. Similarly, a cross-section through the back-arc spreading center above the Tonga subduction zone (Fig. 3.7-20) shows that Q_α exceeds 10,000 within the cold and rigid subducting slab, but is less than 75 beneath the hot back-arc basin. Such attenuation data, especially when combined with velocity data, are valuable for tectonic studies.

3.8 Composition of the mantle and the core

Seismology yields information about velocities within the earth. To derive inferences about the composition of the earth, the seismological data are combined with results from geology, geodesy, geomagnetism, cosmochemistry, and the physics and